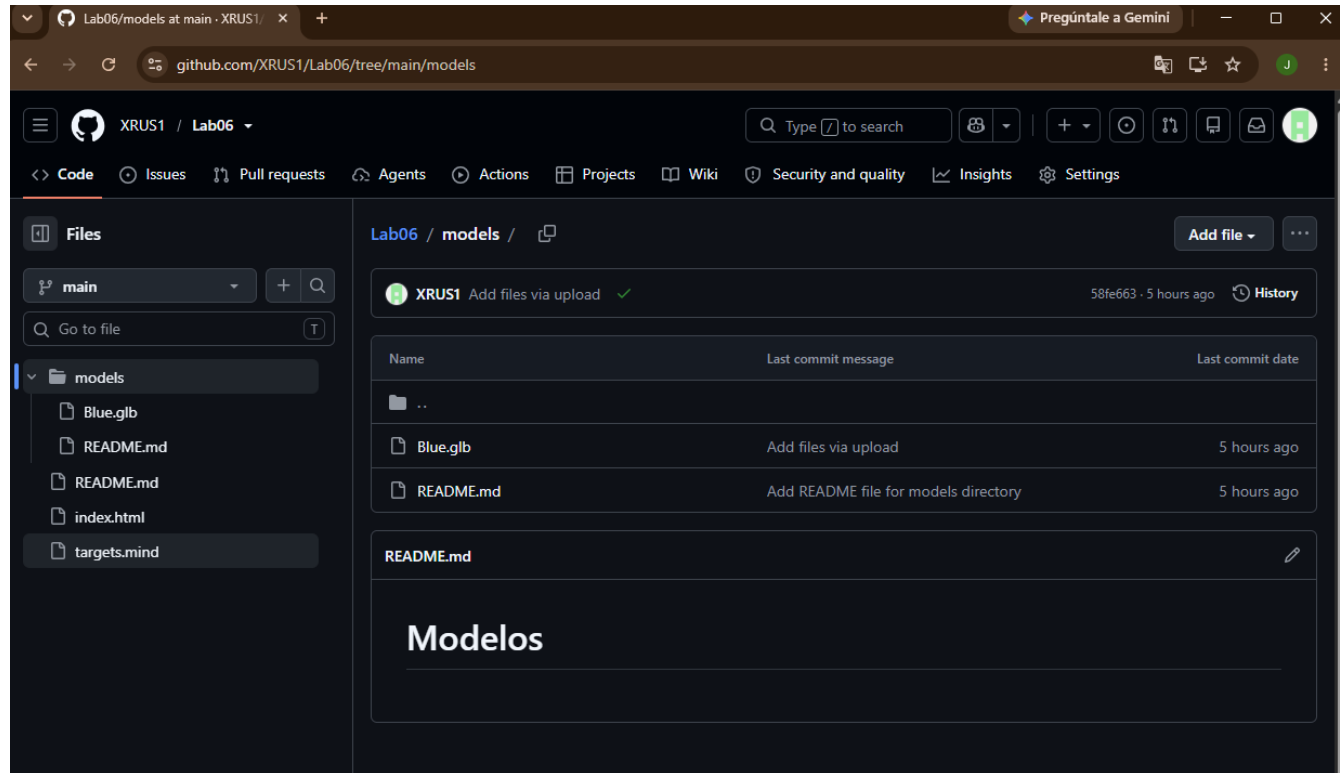


Folders Github



index.html

```
<!DOCTYPE html>
```

```
<html lang="es">
```

```
<head>
```

```
<meta charset="utf-8">
```

```
<meta name="viewport" content="width=device-width, user-scalable=no, minimum-scale=1.0, maximum-scale=1.0">
```

```
<title>Dragón Azul AR - MindAR</title>
```

```
<style>
```

```
* { box-sizing: border-box; margin: 0; }
```

```
body { touch-action: none; overflow: hidden; width: 100vw; height: 100vh; background: #000; }
```

```
#ui { position: absolute; top: 0; left: 0; z-index: 10; padding: 1.2em; max-width: 400px; background:
rgba(0,0,0,0.82); backdrop-filter: blur(10px); border-radius: 0 0 16px 0; font-family: sans-serif; }

h1 { font-size: 1.15em; color: #50a8fc; margin-bottom: 0.4em; }

p { font-size: 0.8em; color: #aaa; margin-bottom: 1em; }

#status { margin-top: 0.8em; font-size: 0.85em; color: #ccc; min-height: 1.4em; }

#status.ok { color: #51cf66; }

#status.err { color: #ff6b6b; }

#status.info { color: #74c0fc; }

#debug { margin-top: 1em; font-size: 0.7em; color: #888; max-height: 200px; overflow-y: auto; }

</style>
```

```
<script async src="https://unpkg.com/es-module-shims@1.3.6/dist/es-module-shims.js"></script>
<script type="importmap">
{
  "imports": {
    "three": "https://unpkg.com/three@0.160.0/build/three.module.js",
    "three/addons/": "https://unpkg.com/three@0.160.0/examples/jsm/",
    "mindar-image-three": "https://cdn.jsdelivr.net/npm/mind-ar@1.2.2/dist/mindar-image-three.prod.js"
  }
}
</script>
</head>
<body>

<div id="ui">
  <h1> 🐉 Dragón Azul AR</h1>
  <p>Apunta la cámara a la tarjeta.</p>
  <div id="status">Iniciando...</div>
  <div id="debug"></div>
</div>
```

```

<script type="module">
import * as THREE from 'three';
import { GLTFLoader } from 'three/addons/loaders/GLTFLoader.js';
import { MindARThree } from 'mindar-image-three';

const statusEl = document.getElementById('status');
const debugEl = document.getElementById('debug');

function log(msg) {
  console.log(msg);
  debugEl.innerHTML += msg + '<br>';
  debugEl.scrollTop = debugEl.scrollHeight;
}

// Verificar targets.mind
log('Verificando targets.mind...');
fetch('./targets.mind')
  .then(r => {
    if (!r.ok) throw new Error('Status: ' + r.status);
    log('✅ targets.mind OK (' + r.status + ')');
    iniciarMindAR();
  })
  .catch(err => {
    log('❌ Error targets.mind: ' + err.message);
    statusEl.textContent = '❌ Error: targets.mind no encontrado';
    statusEl.className = 'err';
  });

async function iniciarMindAR() {

```

```
try {  
  log('Creando MindARThree...');  
  
  const mindarThree = new MindARThree({  
    container: document.body,  
    imageTargetSrc: './targets.mind'  
  });  
  
  const { renderer, scene, camera } = mindarThree;  
  
  // Color space para modelos GLB modernos  
  renderer.outputColorSpace = THREE.SRGBColorSpace;  
  renderer.toneMapping = THREE.ACESFilmicToneMapping;  
  
  log('✅ MindARThree creado');  
  
  // Luces (mismas que tu código WebXR)  
  scene.add(new THREE.HemisphereLight(0xa5a5a5, 0x898989, 3));  
  const dirLight = new THREE.DirectionalLight(0xffffff, 3);  
  dirLight.position.set(1, 1, 1).normalize();  
  scene.add(dirLight);  
  log('✅ Luces añadidas');  
  
  // Crear anchor  
  log('Creando anchor...');  
  const anchor = mindarThree.addAnchor(0);  
  
  // Helper visual: ejes de 10cm para debuggear posición del anchor  
  const axesHelper = new THREE.AxesHelper(0.1);  
  anchor.group.add(axesHelper);
```

```
log('✅ Anchor + ejes de debug creados');
```

```
// Eventos del anchor
```

```
anchor.onTargetFound = () => {
```

```
  statusEl.textContent = '🎯 ¡Dragón detectado!';
```

```
  statusEl.className = 'ok';
```

```
  log('🎯 Target FOUND');
```

```
};
```

```
anchor.onTargetLost = () => {
```

```
  statusEl.textContent = '👁️ Buscando...';
```

```
  statusEl.className = 'info';
```

```
  log('👁️ Target LOST');
```

```
};
```

```
// Cargar modelo Blue.glb (misma lógica que tu código WebXR)
```

```
log('Cargando Blue.glb...');
```

```
const loader = new GLTFLoader();
```

```
loader.load('./models/Blue.glb', (gltf) => {
```

```
  log('✅ Modelo cargado. Animaciones: ' + gltf.animations.length);
```

```
// ==== MISMA CONFIGURACIÓN QUE TU CÓDIGO WEBXR ====
```

```
const container = new THREE.Group();
```

```
const model = gltf.scene;
```

```
// Escala idéntica a tu programa que funciona
```

```
container.scale.set(1.5, 1.5, 1.5);
```

```
container.rotation.y = 0;
```

```

// Centrar modelo (idéntico a tu código WebXR)
const box = new THREE.Box3().setFromObject(model);
const center = box.getCenter(new THREE.Vector3());
model.position.set(-center.x, -center.y, -center.z);

container.add(model);

// En MindAR, el dragón debe estar SOBRE la tarjeta (z positivo sale hacia la cámara)
// y ligeramente elevado para no estar enterrado en el plano
container.position.set(0, 0.05, 0.05);

anchor.group.add(container);
log('✅ Modelo añadido al anchor. Centro: ' +
    center.x.toFixed(2) + ',' + center.y.toFixed(2) + ',' + center.z.toFixed(2));

// Animación: índice 4 (FlyIdle) - idéntico a tu código WebXR
let mixer = null;
if (gltf.animations && gltf.animations.length > 0) {
    mixer = new THREE.AnimationMixer(model);
    const indice = 4; // FlyIdle
    const clip = gltf.animations[indice] || gltf.animations[0];
    log('🎬 Animación [' + indice + ']: ' + clip.name);
    mixer.clipAction(clip).play();
}

// Loop de animación OFICIAL MindAR
const clock = new THREE.Clock();
renderer.setAnimationLoop(() => {
    if (mixer) mixer.update(clock.getDelta());
    renderer.render(scene, camera);
});

```

```
});
```

```
log('✅ Loop iniciado');
```

```
statusEl.textContent = '✅ Listo. Muestra tarjeta.';
```

```
statusEl.className = 'ok';
```

```
},
```

```
(xhr) => {
```

```
  if (xhr.total > 0) {
```

```
    const pct = Math.round((xhr.loaded / xhr.total) * 100);
```

```
    if (pct % 20 === 0) log('⌚ Modelo: ' + pct + '%');
```

```
  }
```

```
},
```

```
(err) => {
```

```
  log('❌ Error modelo: ' + err.message);
```

```
  statusEl.textContent = '❌ Error al cargar modelo';
```

```
  statusEl.className = 'err';
```

```
  console.error(err);
```

```
});
```

```
// Iniciar cámara (DESPUÉS de toda la configuración)
```

```
log('Iniciando cámara...');
```

```
await mindarThree.start();
```

```
log('✅ Cámara iniciada');
```

```
statusEl.textContent = '✅ Cámara OK. Muestra tarjeta.';
```

```
} catch (err) {
```

```
  log('❌ Error general: ' + err.message);
```

```
  statusEl.textContent = '❌ Error: ' + err.message;
```

```
  statusEl.className = 'err';
```

```
    console.error(err);  
  }  
}  
</script>  
  
</body>  
</html>
```

Primero Muestra el QR y después la tarjeta

Código QR



Tarjeta

